

Unit

2

Lesson

1



Innovators in action.

Plan Your Perfect Screen!

Time Frame:[90 min].

Ready, Set, Learn!

By the end of this lesson students will be able to:

- Design the user interface of a mobile app using MIT App Inventor.
- Understand and apply basic accessibility principles to make my app helpful for users with visual impairments.
- Plan and organize the layout of the app to ensure accessibility and ease of use.

Challenge questions of the lesson.

1. How did you design your app's screen to make it easier for someone with visual impairments to use?
2. What features did you include in your app's layout to make it simple and clear to navigate?
3. Why is it important to think about accessibility when designing an app? Who benefits from it?
4. Which components (buttons, labels, sounds, etc.) did you choose for your app's user interface, and why?
5. If you had to improve your app's design for someone who cannot see the screen at all, what would you add or change?

SEL Relation

- Empathy & Perspective-Taking:
 - By designing an app for users with visual impairments, students are encouraged to consider the needs, feelings, and challenges of others. They begin to think like inclusive designers—understanding that not everyone experiences technology the same way.
- Responsible Decision-Making:
 - As students make choices about color, text size, layout, and sound, they practice making thoughtful decisions that have real impacts on user experience. They learn to balance creativity with responsibility and user needs.
- Self-Awareness & Growth Mindset:
 - Students reflect on how their design choices affect others, and they revise their work based on feedback and testing. This builds a growth mindset, as they learn that improvement and iteration are part of the design process.

Engage



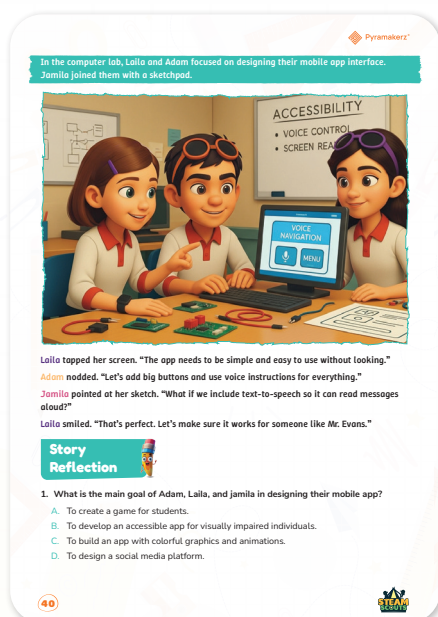
Teacher Role:

- Read the scene aloud or guide students through reading it together to activate empathy and context.
- Pause to emphasize key dialogue that shows the team's design thinking (e.g., "Let's add big buttons and voice instructions").
- Highlight how the characters are considering users with visual impairments and introduce the term "accessibility."
- Guide a short discussion using the multiple-choice questions, prompting students to justify their answers based on the dialogue.
- Use the Think prompt to transition students into imagining themselves as designers—encouraging creativity, empathy, and open-ended thinking.
- Offer drawing support or brainstorming tools to help students visualize their accessible app ideas.

Expected From Students:

- Actively listen or read the story and identify the team's goal and ideas for accessibility.
- Participate in answering the reflection questions and explaining their reasoning.
- Think deeply about challenges faced by users who are visually impaired.
- Respond to the prompt: imagine and describe (or draw) features they would include in a helpful, inclusive app.
- Begin developing an empathetic mindset by placing themselves in the shoes of the user.

Parts of the books included.



2. Adam proposed adding a text-to-speech function. What problem does this solve?
- A. It allows the app to read text out loud for visually impaired users.
 - B. It makes the app more fun by adding robotic voices.
 - C. It helps users type faster.
 - D. It replaces the need for buttons in the app.



Let's Think!

If you were designing an app for people who have trouble seeing, what features would you include?

Think about how the app should look, how people will use it, and what special tools or buttons could make it easier for them to use. "Draw it if you want"

Story



1. Begin by gathering students and saying:

- “Today we’re going to step into the shoes of real designers. You’ll meet a team of students—Laila, Adam, and Jamila—who are working on something important. Let’s read what they’re doing and see if we can figure out the challenge they’re solving.”
- Then, either read the scene aloud with voice expression or assign character roles and have students perform it as a short script.
- After the reading, pause and ask:
 - “Who is Mr. Evans? Why do you think they’re designing the app to help him?”

Possible Strategies to Implement:

1. Visible Thinking Routine – “Step Inside”

- Ask students:
 - “Step into Jamila’s shoes. What is she thinking when she suggests adding text-to-speech?”
 - “What do you think Mr. Evans might need from this app?”
 - This builds empathy and perspective-taking.

2. Think-Pair-Share

- Prompt:
 - “What did you hear that tells you they’re designing for someone who has trouble seeing?”
 - Have students think quietly, discuss with a partner, then share with the class.

3. Four Corners (Opinion-Based Movement).

- Post these on 4 classroom corners:
 - Big Buttons Help
 - Voice Instructions Help
 - Text-to-Speech Helps
 - All Are Important
- Ask: “Which feature do you think is most helpful for someone like Mr. Evans? Go to the corner that matches your answer and be ready to explain why.”

4. Anchor Chart – “Designing for Accessibility”

- Create a live class chart titled “What Makes Apps Accessible?”
- Start filling it with student ideas as they reflect on the scene. Add examples like big buttons, voice tools, or screen readers.

Story Reflection



Objective:

Students will demonstrate understanding of the purpose and design considerations of an accessible app interface by identifying key goals, design features, and their relevance to users with visual impairments.

Answers:

1. What is the main goal of Adam, Laila, and Jamila in designing their mobile app?

A. To develop an accessible app for visually impaired individuals.

2. Adam proposed adding a text-to-speech function. What problem does this solve?

A. It allows the app to read text out loud for visually impaired users.

Possible Strategies for Implementation:

1. Think-Pair-Share:

- Before revealing the answer, have students think individually, then pair up to justify their choices. Ask:
 - “Why do you think they made that design choice?”
 - “How does this help someone like Mr. Evans?”

2. Evidence Hunt:

- Have students highlight or underline phrases in the story scene that support each correct answer. This builds close reading skills and evidence-based thinking.

3. "Why Not?" Strategy:

- For each incorrect answer choice, ask students why it doesn't fit the story. This encourages deeper comprehension and critical analysis.

4. Anchor Chart Addition:

- After answering, add the design elements from the questions (text-to-speech, large buttons, simple layout) to an anchor chart titled: “Features That Make Apps Accessible.”

5. Role Reversal:

- Ask students:
 - “If you were Mr. Evans, which feature would help you the most? Why?”
 - Let them answer from the user's perspective to build empathy and deeper understanding.

Let's Think!



Objective:

Students will apply empathy and design thinking to imagine app features that support users with visual impairments. They will generate ideas that reflect accessibility principles such as large buttons, voice commands, and text-to-speech features.

Answers (Possible Student Ideas):

- Big, colorful buttons with clear labels.
- Voice instructions or sound feedback.
- Text-to-speech feature to read messages.
- High-contrast color themes (e.g., white text on black).
- Vibration feedback when pressing a button.
- Simple layout with fewer distractions.
- A screen reader mode.
- Zoom-in or magnifier options.



Possible Strategies for Implementation:

1. Empathy Mapping:

- Ask students to imagine they are Mr. Evans. What would they need to use an app easily? What would be hard? Have them write or draw their ideas.

2. Sketch First, Then Share:

- Give time for individual sketching or labeling app ideas. Then invite volunteers to present or explain their design features to a partner or group.

3. Design Gallery Walk:

- Post student drawings or notes around the room. Students walk around, observe ideas, and add sticky notes with compliments or suggestions.

4. Prompt Starters:

- Offer simple prompts to support all learners:
 - "I would add a button that..."
 - "My app would speak when..."
 - "This tool helps people who..."

5. Compare with Real Examples:

- Show a simple example of an existing accessibility feature (e.g., phone with screen reader or high contrast mode) after the activity to help students connect their designs with real-world tools

Explorer's Toolkit



Teacher Role:

- Prepare students for the activity by introducing the idea of accessibility and the challenges faced by visually impaired users.
- Ensure a safe space by setting behavior expectations, especially while students are blindfolded.
- Provide blindfolds and sample app layout printouts or digital mockups.
- Monitor each pair during the activity, offering guidance on how to give clearer verbal instructions.
- Lead a reflective discussion afterward:
 - “What made this difficult? How can good design help solve these problems?”

Expected From Students:

- Actively participate in partner roles: one student as the navigator, the other as the verbal guide.
- Follow safety rules and take turns respectfully.
- Use only spoken instructions to explore the interface, listening and responding thoughtfully.
- Reflect on challenges faced while blindfolded and suggest real improvements that could help visually impaired users.
- Record their observations in writing.

Parts of the books included.

Explorer's Toolkit



DIY Lab

In this activity, you will explore how people with vision impairments use mobile apps and how to make them easier to use. Here's what you'll do:

- Pair up with a classmate—one person will wear a blindfold.
- The partner will describe the layout of a mobile app interface.
- The blindfolded student will navigate the app using only the verbal instructions given.

What was difficult? How could the design be improved?

STEAM SCOUTS



Figure 18: Exploring App Accessibility for the Visually Impaired.

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DIY Lab

Objective:

- Students will experience firsthand the challenges of navigating a digital interface without vision, and reflect on how accessible design choices (such as voice guidance, button size, and layout simplicity) support inclusive technology use.

Materials Needed:

- Blindfolds (or paper towels/fabric strips).
- App layout printouts or wireframe sketches.
- Pencil or pen.
- Student worksheet or notebook for reflection.
- Safe open space for pairs to work.

Activity Instructions:

1. Pair up students and assign roles: one will be blindfolded, the other will be the guide.
2. The guide describes the layout of a basic mobile app screen using only spoken words.
3. The blindfolded student tries to find or "tap" the correct button or navigate based only on those instructions.
4. After switching roles, both students discuss:
 - "What was hard? What helped? What would make this easier for someone who cannot see?"
5. Students write or draw improvements that could help someone navigate better.

Possible Strategies for Implementation:

1. Think Aloud Protocol:

- Encourage the guide to verbalize everything they see and why it matters:
 - "There's a large yellow button at the bottom labeled 'CALL' – it's to the right of a smaller message icon."

2. Empathy Pause:

- After the blindfolded student navigates, pause and ask:
 - "How did that feel?" "What would help make this easier for someone every day?"

3. Role Reversal:

- Let students experience both roles so they understand the importance of clear design and clear communication.

4. Reflective Drawing:

- After the activity, students sketch a redesigned app interface with accessibility improvements based on what they experienced.

5. Group Share-Out:

- Invite a few students to share their biggest design insight with the class

Explain



Teacher Role:

- Introduce the video by explaining that students will explore real features that improve mobile apps for users with visual impairments.
- Before playing the video, ask students to listen for specific features such as voice commands, text-to-speech, and layout tips.
- Pause the video at key moments to ask guiding questions like:
 - “Why would that feature help someone who can’t see well?”
 - “Have you seen or used a similar feature before?”
- Clarify any unfamiliar vocabulary (e.g., contrast, magnification, accessibility settings).
- After the video, lead a quick recap or discussion on the most important takeaways and how students might include those ideas in their own designs.

Expected From Students:

- Watch the video attentively and take note of 3–4 accessibility features they find interesting or useful.
- Reflect on how each feature helps someone with a visual impairment.
- Participate in class discussion or answer post-video questions about how these features could be used in their app project.
- Use the ideas from the video to guide their design decisions during the Builder or Mastermind task.

Parts of the books included.



Screen Lessons

Click the link to discover which features you can include in your mobile app design.

<https://www.youtube.com/watch?v=DTtTFb4ADJo>



Screen Lessons

Objective:

- Identify key accessibility features used in mobile app design to support users with visual impairments.
- Understand how design choices like layout, contrast, and voice tools improve usability.
- Apply observed features to improve student-designed app interfaces.

Video Link:

<https://www.youtube.com/watch?v=DTtTFb4ADJo>



Possible Strategies for Implementation:

1. Pre-Watching: Set a Purpose.

- Ask students:
 - “If you couldn’t see the screen clearly, what features would you want in an app?”
- Write their ideas on the board to compare later with the video content.
- Share a viewing goal: “While watching, find at least 3 features that help users with visual impairments.”

2. During Watching: Active Engagement.

- Pause at key points to highlight specific features like:
 - Screen readers
 - Voice commands
 - High-contrast themes
 - Zoom or magnifiers
- Use prompts such as:
 - “Why do you think this feature matters?”
 - “How would this help someone like Mr. Evans?”

3. Post-Watching: Discussion and Reflection

- Facilitate a whole-class discussion using prompts like:
 - “Which feature would you add to your app first and why?”
 - “Which ideas from the video match what you already knew? Which ones surprised you?”
- Optionally, have students list or draw 3 features from the video they plan to use in their own app designs.

Elaborate



Teacher Role:

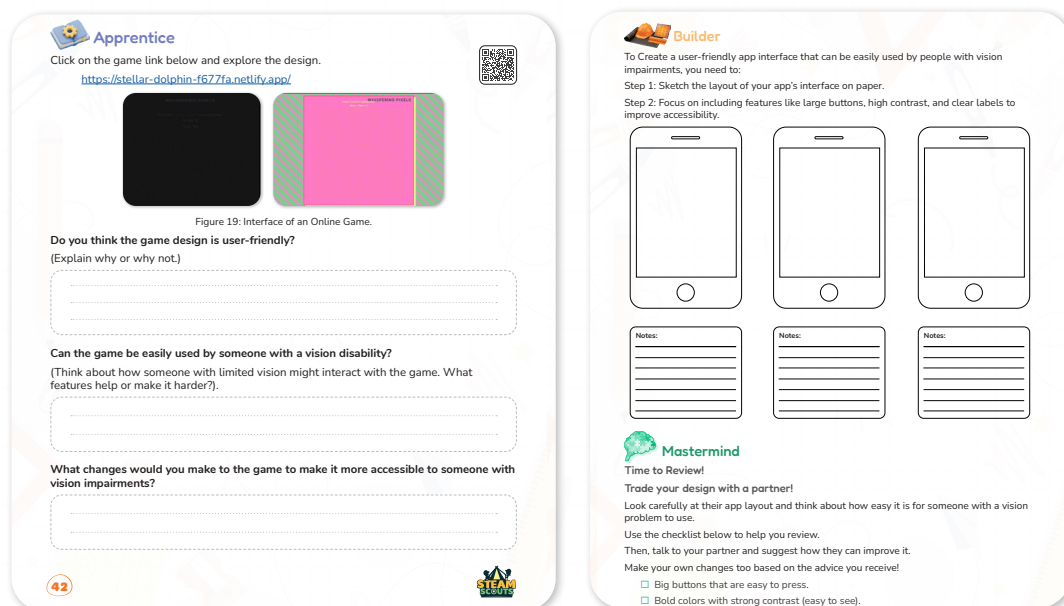
- Guide students through a critical analysis of an existing digital interface by prompting reflection on accessibility and user experience.
- Provide structured time for students to design their own mobile app layout with accessibility in mind.
- Encourage them to apply what they’ve learned from the video, class discussions, and DIY activity.

- Facilitate peer review sessions where students give and receive feedback using a structured checklist focused on inclusive design.
- Prompt students with questions like:
 - “What makes this design easier or harder for someone who can’t see well?”
 - “How could you adjust this layout to be more accessible?”

Expected From Students:

- Evaluate an existing game interface and reflect on whether it is user-friendly for someone with vision challenges.
- Identify design barriers and suggest improvements that support people with visual impairments.
- Sketch a custom app interface that includes accessibility features such as large buttons, voice guidance, bold color contrast, and clear labels.
- Review a peer’s design using a structured checklist and provide constructive feedback.
- Revise their own design based on the feedback they receive, demonstrating understanding of inclusive and user-centered design principles.

Parts of the books included.



Apprentice

Objective:

Students will evaluate the accessibility of a digital game interface and propose improvements based on the needs of users with visual impairments.

Activity Link: <https://gemini.google.com/app/ad56b285f546cdad>



Answers:

- A. Do you think the game design is user-friendly?
- No, it is not very user-friendly because the text is small and the screen is very dark. It's hard to see where to click or what to do next.

- B. Can the game be easily used by someone with a vision disability?
- o No, someone with a vision problem might struggle because the colors are too dark and there are no voice instructions. There's not enough contrast or large buttons.
- C. What changes would you make to the game?
- o I would add high-contrast colors, bigger text, audio instructions, and maybe a screen reader mode to make it easier for someone with limited vision.

Activity Instructions:

1. Share the game link with the class. Make sure all students can access it individually or in pairs.
2. Ask students to explore the game interface and observe the design, layout, text, and colors.
3. Instruct students to reflect on:
 - o How easy or hard it is to navigate.
 - o Whether someone with a vision impairment could use it easily.
 - o What could be improved for accessibility.
4. Have students write their answers directly in the worksheet or in their journals.
5. Review student responses to assess their understanding of accessibility design principles.

Possible Strategies for Implementation:

1. Compare & Contrast:

- After exploring the game, show or reference an example of a well-designed accessible app. Ask students to compare both.

2. Think-Pair-Share:

- Let students discuss their answers in pairs before writing to clarify and expand their thinking.

3. Anchor Chart: "What Makes a Game Accessible?"

- Build a live chart using student answers to define best practices for accessible design.

4. Checklist Prompt:

- Offer a simplified checklist to help students assess the game:
 - o Are the buttons big?
 - o Is the text readable?
 - o Is there sound or voice guidance?
 - o Is the layout clear?



Objective:

Students will apply their understanding of accessible design principles by sketching a mobile app layout that includes features such as large buttons, high contrast, voice features, and simple navigation. The goal is to create an interface usable by people with vision impairments.

Materials Needed:

- Phone screen templates.
- Pencils, markers, or crayons (especially bold colors).
- Ruler (optional, for neat layout).

Activity Instructions:

1. Remind students what makes an app interface accessible (big buttons, clear labels, contrast, voice cues).
2. Distribute the sketch templates or provide plain paper.
3. Guide students through:
 - **Step 1:** Sketching the layout of a mobile app interface from their own idea.
 - **Step 2:** Labeling parts of the screen and incorporating accessibility features.
 - Examples: a large “Read Message” button, a magnifier icon, or text-to-speech toggle.
4. Encourage students to focus on function before decoration—design with the user in mind.
5. Once finished, prompt students to prepare for peer review by thinking:
 - “Would someone with low vision understand what to do on this screen?”

Possible Strategies for Implementation:

1. Design Thinking Prompt:

- Ask: “If Mr. Evans were using your app, where would he tap first? What would help him succeed?”

2. Model First:

- Quickly sketch a sample layout on the board with a few accessibility features and explain your thinking.

3. Accessibility Checklist Anchor:

- Display a live checklist of key features students should consider:
 - Large, labeled buttons
 - Clear icon meaning
 - High contrast or bold colors

- Audio/voice features
- Simple layout (not cluttered)

4. Gallery Walk Option:

- After sketching, display the designs around the room and have students do a silent walk to observe and jot down inspiration.



Mastermind

Objective:

Students will review and evaluate a peer's app interface sketch using a structured accessibility checklist. They will provide feedback based on inclusive design principles and revise their own work accordingly.

Materials Needed:

- Student sketches of app layouts (from the Builder activity).
- peer review checklists.
- Pencils or pens for annotation.
- Optional: colored markers for highlighting feedback.

Activity Instructions:

1. Have students pair up and exchange their app sketches.
2. Ask each student to carefully review their partner's layout, using the checklist to mark which features are present.
3. Encourage students to write comments or suggestions next to their partner's sketch or on a sticky note.
4. After reviewing, students should have a brief discussion with their partner to explain their suggestions respectfully.
5. Each student will then revise their own design, making improvements based on the feedback they received.

Possible Strategies for Implementation:

1. Model the Process:

- Demonstrate how to give kind, specific, and helpful feedback using a sample sketch.

2. Checklist Reflection:

- After the review, ask students to reflect:
- "Which box on the checklist was hardest to meet in your design? Why?"

3. Two Stars and a Wish:

- Guide students to give feedback using this format:
 - One thing I like...

- o Another strong point is...
- o One thing I'd improve is...

4. Traffic Light Feedback:

- Provide red/yellow/green dots or sticky notes:
 - o Green = clear and helpful.
 - o Yellow = could use some improvement.
 - o Red = confusing or missing feature.

Evaluate



Teacher Role:

- Guide students in translating their paper-based UI designs into functioning screens in MIT App Inventor.
- Support them in identifying the correct components (buttons, labels, color blocks, etc.) and placing them according to accessibility principles.
- Reinforce the connection between their original sketches and their digital design decisions.
- Encourage iteration and refinement, especially based on checklist feedback from peer reviews.
- Offer technical assistance and praise effort, creativity, and problem-solving.

Expected From Students:

- Use their sketched app layout as a reference to create a digital version in MIT App Inventor.
- Select and arrange components that make the app accessible to users with vision impairments (e.g., big buttons, bold text, clear contrast).
- Apply the features they've learned about throughout the lesson to improve their app's usability.
- Watch the tutorial video when needed to support independent building.
- Submit a functional screen design and reflect on how it helps users with visual challenges.

Parts of the books included.



Showcase

Time to Bring Your Sketches to Life!

Now that you've planned your app interface, it's time to start building it using MIT App Inventor.

Use your sketch as a guide and begin designing your app on the screen!

Need a little help?

Watch this video to guide you while you design your app using App Inventor!

https://www.youtube.com/watch?time_continue=59&v=ORpFBVDFC74&embeds_referring_euri=https%3A%2F%2Fchatgpt.com%2F&source_ve_path=MTM5MTE3LDEzOTExNywyMzg1MQ



Figure 20: App Interface in MIT App Inventor.



Showcase

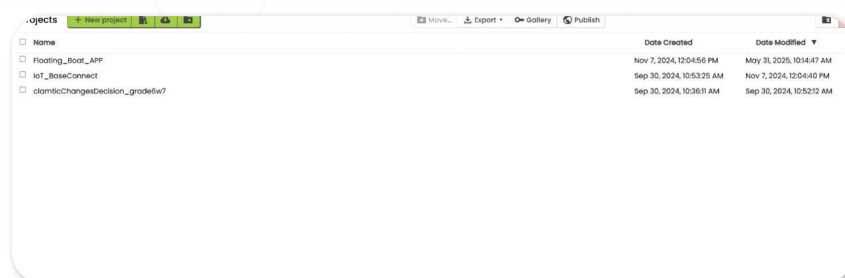
Objective:

Students will apply their knowledge of accessible design and user interface principles by building a functional app screen in MIT App Inventor that supports users with visual impairments.

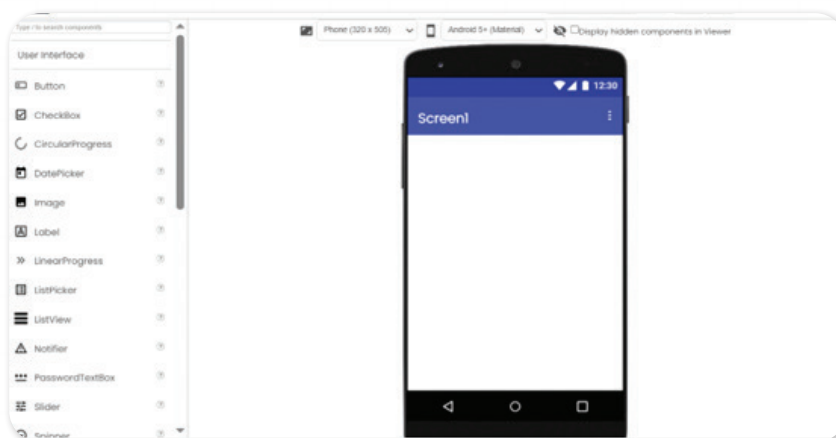
Ready Made Mobile app: <https://drive.google.com/file/d/1k-LZCwAtxzdSD25kWNZQhYKEnfdsXXE/view>

Activity Instructions:

1. Open MIT App Inventor and create a new project.



2. Use your paper sketch to begin laying out the app screen.
3. Add components such as buttons, labels, and speech elements.



4. Follow your accessibility checklist (big buttons, bold colors, etc.) while placing items on the screen.

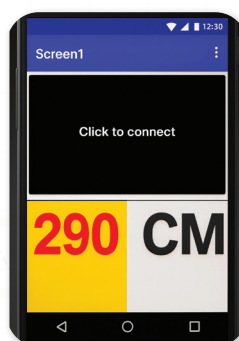


Figure 20: App Interface in MIT App Inventor.

5. Watch the video tutorial if you need help locating or modifying components.

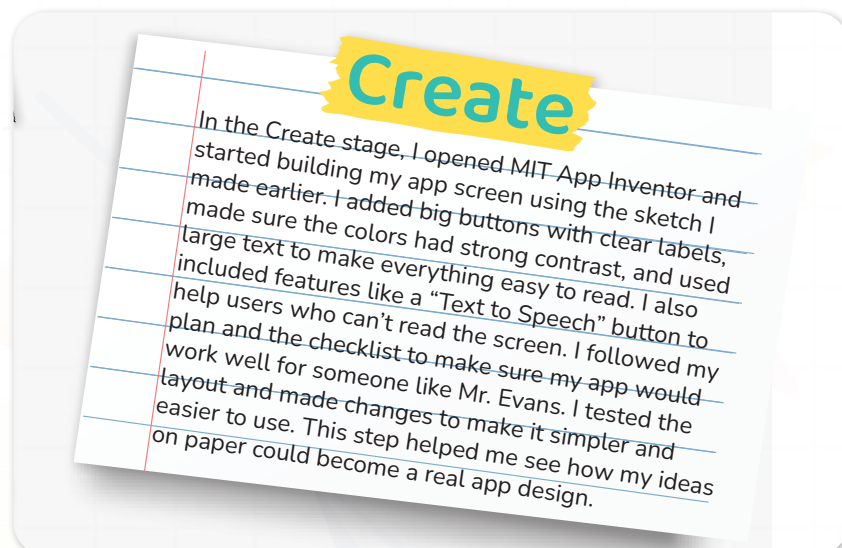
https://www.youtube.com/watch?time_continue=59&v=ORpFBVDFC74&embeds_referring_euri=https%3A%2F%2Fchatgpt.com%2F&source_ve_path=MTM5MTE3LDEzOTExNywyMzg1MQ



Showcase Rubric:

Criteria	Excellent (4 Point)	Good (3 Points)	Developing (2 Points)	Needs Work (1 Point)
Accessible Layout Design (10 pts).	Clear, well-spaced layout with accessibility top of mind.	Mostly clear layout with basic accessibility.	Some layout issues, limited accessibility features.	Cluttered or confusing layout, missing accessibility.
Component Selection (10 pts).	Uses appropriate UI components (buttons, labels, icons) effectively.	Uses most key components appropriately.	Missing some components or misused.	Few or no relevant components used.
Visual Contrast & Text Size (10 pts).	Strong color contrast and easy-to-read labels.	Mostly readable, some color or text issues.	Hard-to-read text or low contrast.	Text not readable or poorly placed.
Voice or Accessibility Feature (10 pts).	Includes TTS, sound cues, or visual feedback.	Includes basic accessibility elements.	Includes only 1 accessibility feature.	No accessibility features included.
Effort & Completion (10 pts).	Fully built, neat, thoughtful design.	Complete with minor gaps.	Partial design or minimal effort.	Incomplete or off-task.

It's time to move to the Create part and add this.





Now I Can

Objective:

Students will reflect on their learning by identifying what they can now do as a result of the lesson, including designing a mobile app interface in MIT App Inventor and applying accessibility principles to support users with visual impairments.

Activity Instructions:

2. Display the “Now I Can” box on the board or distribute it as a student handout.
3. Read each statement aloud and ask students to reflect silently:
 - “Can I do this confidently now? Which one was hardest for me? Which one was most fun?”
4. Have students:
 - Tick or highlight each “Now I Can” statement they feel confident about.
 - Write a short sentence or draw a small icon next to the one they’re most proud of.
 - Optionally complete the sentence:
“One thing I can do now that I couldn’t do before is...”
5. Facilitate a brief whole-class or small-group sharing session to celebrate learning progress and surface any lingering questions.

Relation to Standards

1. ISTE Standards for Students:

- **1.1 Empowered Learner**
 - Students take ownership of their learning by using digital tools (MIT App Inventor) to express their ideas and build accessible technology.
- **1.4 Innovative Designer**
 - Students use a design process to develop creative solutions that address real-world needs—specifically, designing an app for users with visual impairments.
- **1.5 Computational Thinker**
 - Students apply logic and structured thinking to organize their app interface and consider accessibility in layout and function.

2. Next Generation Science Standards (NGSS):

- **ETS1.A: Defining and Delimiting Engineering Problems**
 - Students identify the challenge of creating a user interface for visually impaired users and define specific design goals.
- **ETS1.B: Developing Possible Solutions**
 - Through sketching, peer feedback, and prototyping with App Inventor, students evaluate and improve their solutions.

3. Common Core State Standards – ELA (CCSS.ELA-LITERACY):

- **CCSS.ELA-LITERACY.SL.5.1**
 - Students engage in collaborative discussions (during peer review and critique) to clarify ideas and express opinions respectfully.
- **CCSS.ELA-LITERACY.W.5.2**
 - Students explain their design ideas and decisions clearly in written reflections and EDP journal entries.

4. UNESCO ICT Competency Framework for Teachers:

- Module 2: Knowledge Deepening – Problem-Solving with Technology.
- Students use digital design tools (App Inventor) to solve authentic problems and create inclusive technological solutions.

5. Sustainable Development Goals (SDGs):

- **SDG 4: Quality Education.**
 - Promotes inclusive learning by emphasizing accessibility and empathy through app design.
- **SDG 10: Reduced Inequalities.**
 - Encourages students to consider how technology can remove barriers for people with disabilities and foster digital inclusion.